

PRISMA-ScR Checklist

Supplementary Material S1

Artificial Intelligence-Driven Forecasting of Relief Demand in Emergency Supply Chains: A PRISMA-ScR Scoping Review of Research Attention and Evidence of Efficacy

Manuscript SASC-D-26-00151, *Systems and Soft Computing*

Completed against the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR), Tricco et al., *Annals of Internal Medicine* 169(7):467–473, 2018. Item numbering follows the published checklist. Section numbers refer to the revised manuscript.

#	Item	Checklist description (abridged)	Reported in
TITLE			
1	Title	Identify the report as a scoping review.	Title: "... A PRISMA-ScR Scoping Review..."
ABSTRACT			
2	Structured summary	Provide a structured summary including background, objectives, eligibility criteria, sources of evidence, charting methods, results and conclusions.	Abstract. States the design, the six databases, the final query date, the eligibility outcome ($N = 113$), the charting convention (multi-label), the principal results and the conclusion.
INTRODUCTION			
3	Rationale	Describe the rationale in the context of what is already known; explain why the questions lend themselves to a scoping review.	Section 1 and Section 1.1; Section 2 positions the review against five existing reviews and states the three respects in which it differs.
4	Objectives	State the questions and objectives; state the key elements (e.g. population, concept, context).	Section 1.3 (RQ1–RQ4); PCC elements in Section 3.2.
METHODS			
5	Protocol and registration	Indicate whether a protocol exists, where it can be accessed, and registration information.	Section 3.1. No protocol was registered; PROSPERO does not accept scoping reviews. The methodological framework followed (Arksey & O'Malley; Levac et al.; JBI) is named.
6	Eligibility criteria	Specify characteristics of the sources of evidence used as eligibility criteria, including years considered and language, and provide a rationale.	Section 3.2. Four numbered inclusion and four numbered exclusion criteria; publication window inception to 22 August 2026; English language; peer-reviewed sources only, with the exclusion of grey literature justified.
7	Information sources	Describe all information sources, including dates of coverage and the date the most recent search was executed.	Section 3.3 and Table 1. Six databases, each searched from inception; final query August 2026. Citation chasing described.
8	Search	Present the full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Table 1 gives the complete Boolean string as executed for <i>all six</i> databases, with the fields searched and the records returned.
9	Selection of sources of evidence	State the process for selecting sources of evidence.	Section 3.4 and Figure 2. Duplicate screening by two reviewers, adjudication by a third; counts at every stage; itemised full-text exclusion reasons.
10	Data charting process	Describe the methods of charting data, whether the process was performed independently or in duplicate, and any processes for obtaining and confirming data from investigators.	Section 3.5 and Figure 3. Form piloted on ten studies, then applied in duplicate with adjudication; derived variables produced by documented rules.
11	Data items	List and define all variables for which data were sought and any assumptions and simplifications made.	Section 3.5, Figure 3 and Table 2 (rules R1–R6). Assumptions: multi-label method family, mutually exclusive remaining variables, exclusion of missing values from denominators, are stated in Section 3.5.1.

#	Item	Checklist description (abridged)	Reported in
12	Critical appraisal of individual sources of evidence	If done, provide a rationale for conducting it; describe the methods used and how the information was used in any data synthesis (<i>optional item</i>).	Section 3.6. No critical appraisal was performed. The choice is declared explicitly, justified against Tricco et al. (2018), Munn et al. (2018) and Pollock et al. (2021), and its two consequences for the synthesis are stated. The instruments that would have applied (PROBAST, CHARMS, TRIPOD+AI, MMAT) are named.
13	Synthesis of results	Describe the methods of handling and summarising the data that were charted.	Section 3.7. Three complementary syntheses: descriptive numerical summary, comparative synthesis by method family and hazard, and evidential synthesis. No meta-analysis, with the reason given.
RESULTS			
14	Selection of sources of evidence	Give numbers of sources screened, assessed for eligibility and included, with reasons for exclusions at each stage, ideally using a flow diagram.	Figure 2 (PRISMA-ScR flow diagram) and Section 3.4. 1,155 records identified, 283 duplicates removed, 872 screened, 748 excluded, 124 full texts assessed, 11 excluded with itemised reasons, 113 included.
15	Characteristics of sources of evidence	For each source, present characteristics for which data were charted and provide the citations.	Table 6 lists all 113 sources on the charted variables; the complete machine-readable record is <code>master_extraction.csv</code> .
16	Critical appraisal within sources of evidence	If done, present data on critical appraisal (<i>optional item</i>).	Not applicable; see item 12. Evidential variables (performance evidence, deployment status) are reported instead, in Section 7 and Figure 15.
17	Results of individual sources of evidence	For each included source, present the relevant data that were charted that relate to the review questions and objectives.	Table 6 (all sources); Table 5 (extracted comparative performance results); Section 5 (narrative synthesis by method family).
18	Synthesis of results	Summarise and/or present the charting results as they relate to the review questions and objectives.	Section 5 (RQ1), Section 6 (RQ2), Section 7 (RQ3), Section 8 (RQ4); Figures 4–15; Tables 3–6.
DISCUSSION			
19	Summary of evidence	Summarise the main results, link to the review questions and objectives, and consider the relevance to key groups.	Section 12; managerial and policy implications for practitioners in Section 10 (M1–M8).
20	Limitations	Discuss the limitations of the scoping review process.	Section 11 (L1–L6), including the absence of critical appraisal, the conservative and retrievability-dependent performance coding, the recoding differences from the earlier version, the language and indexing restriction, the boundary decisions, and the absence of meta-analysis.
21	Conclusions	Provide a general interpretation of the results with respect to the review questions and objectives, and discuss the implications and next steps.	Section 12; research agenda D1–D10 in Section 9.1.
FUNDING			
22	Funding	Describe sources of funding for the included sources of evidence, as well as sources of funding for the scoping review; describe the role of the funders.	Declaration of competing interest and funding statement, end matter. Funding of the included sources was not charted; this is noted as outside the review's scope.

Note on reproducibility. Every result reported against items 14–18 is computed by released code from a single released dataset.